

Making good decisions with counterfactuals

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Abstract

Making a good decision often requires knowing something counterfactual: whether a person would be better off if we acted than if we didn't. Formal work shows how both experimental and observational data ought to be combined to inform good decisions. Here, we ask whether people's intuitions follow this normative standard. In three experiments ($N = 354$), participants decided whether or not to act based on observational and experimental data. Participants performed better than chance but far from optimally. Trial-by-trial feedback about the consequences of their choices, including what would have happened had they chosen otherwise, didn't improve performance. The breakdown was at the inference step: participants' estimates of whether a person would benefit from the treatment, be harmed by it, or be left unaffected were almost unrelated to normative predictions. Yet, participants' decisions followed coherently from those (incorrect) estimates. Experts in statistics, economics, and data science made better decisions than novices, but their counterfactual inferences were no more accurate. Counterfactual reasoning is critical for making good decisions from experimental and observational data, but its logic doesn't come naturally to people.

Keywords: counterfactuals; causality; causal inference; decision-making

Corresponding author: Tobias Gerstenberg  <https://orcid.org/0000-0002-9162-0779>; gerstenberg@stanford.edu. All materials, data, analyses, pre-registrations, experiment demos, and an interactive model interface are available at https://github.com/cicl-stanford/good_decisions.

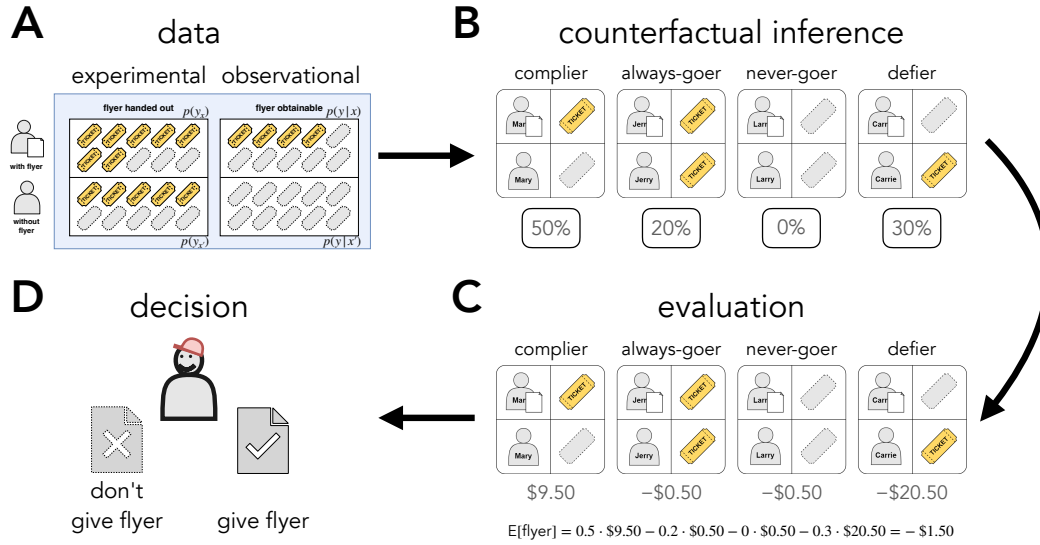
Introduction

The adage “correlation is not causation” is well known. It reminds us to be cautious about data from observational studies. Maybe the only reason that X and Y are correlated is that they are confounded by a common cause C , so it wouldn’t help to change X if one wanted to affect Y . Experimental data is the gold standard in science. Randomized controlled trials (RCTs) solve many of the problems of confounding (Deaton & Cartwright, 2018; Kyriacou & Lewis, 2016), so we often trust experimental data more than observational data. However, sometimes we have both. A doctor may need to decide whether to administer a treatment to a patient, knowing both the results of an RCT and a large-scale observational study (Barton, 2000). A manager who has data from an A/B test and from how people behaved on their own needs to decide whether to launch a new advertisement (Gordon, Zettelmeyer, Bhargava, & Chapsky, 2019). How should one combine experimental and observational data to make good decisions?

In recent years, formal tools have been developed that answer this question (Bareinboim & Pearl, 2016; Colnet et al., 2024; Li & Pearl, 2019; Pearl, 1999; Tian & Pearl, 2000). These tools show that making good decisions requires more than distinguishing correlation from causation – it requires counterfactual reasoning (Li & Pearl, 2022; Mueller & Pearl, 2023). Counterfactual reasoning involves asking “What would have happened if things were different?” questions (Byrne, 2016; Gerstenberg, 2022, 2024; Goodman, 1956; Kahneman & Tversky, 1982). Intuitively, a doctor wants to give a treatment to a patient who would benefit from it and not to one who would get harmed (Dawid & Senn, 2023; Greenland & Robins, 1986; Heckerman & Shachter, 1995; Ritov & Baron, 1990; Sarvet & Stensrud, 2023, 2024). But notice that “benefit” and “harm” are counterfactual constructs (Beckers, Chockler, & Halpern, 2023, 2024; Ben-Michael, Imai, & Jiang, 2024; Bradley & Stefánsson, 2017; Richens, Beard, & Thompson, 2022). A person benefited if they got better *because* of the treatment, meaning they would have been worse off without it. Analogously, a patient was harmed by the treatment if they would have been better off without it. So, ideally, a doctor uses the available experimental and observational data to estimate how likely it is that a given patient would benefit from or be harmed by the treatment, and then makes the optimal decision based on that information.

In this paper, we investigate how both laypeople and experts make decisions based on experimental and observational data. We compare their decisions against the predictions of a normative theory – the counterfactual decision model (CDM) – as well as heuristic models. Across experiments, we manipulate whether participants receive feedback about the quality of their choices, and we ask them to engage with the data in different ways. Overall, we find that both novices and experts largely fail to adhere to the normative standard: they have trouble making the right inferences about the population from the data, leading them to make suboptimal decisions. Experts perform better than novices, though, and novices perform better than chance. We also find that while feedback itself doesn’t improve performance, having participants more closely engage with the data before making a decision helps.

The rest of the paper is organized as follows. We first provide an example of the decision-making problem that participants face in our experiments. We then briefly discuss relevant background literature on the role that causality and counterfactuals play in

**Figure 1**

The counterfactual decision model. We illustrate how the model works in a scenario where we have information on whether movie-goers bought a ticket based on whether they had a flyer or didn't have one. The decision of interest is whether to hand out flyers in this theater in the future. **A** The model takes in experimental and observational data. **B** Based on this data, it computes the probability of four counterfactual types in the population at this theater who differ in how they respond to the treatment. **C** The model assigns a value to each of the different counterfactual types and computes the expected benefit of handing out a flyer versus not handing it out. **D** It uses the expected benefit to decide whether to hand out flyers or not. In this scenario, the expected benefit of handing out the flyer is negative.

decision-making. Afterwards, we introduce the CDM. We compare the predictions of this model, and alternative models, against participants' decisions in three experiments. We conclude by discussing possible reasons why participants' decisions don't accord with the CDM and provide some intuition on how to make better decisions. We highlight limitations of the current work and suggest future research directions.

A summer job at 'Zeus Theater'

Imagine you took a summer job at 'Zeus Theater'. Your job is to determine whether it's profitable to hand out flyers for a new movie. Zeus has already conducted an experimental study and an observational study to help you make an informed decision (see Figure 1A). In the experimental study, one of your colleagues stood outside the movie theater and handed out flyers to a random selection of customers who had just watched a different movie. In the observational study, your colleague didn't hand out the flyers randomly, but instead put up the flyers on display so that each movie-goer could decide themselves whether or not to take one. In both the experimental and observational studies, more people who got the flyer bought a ticket for the advertised movie than people who didn't get the flyer. In fact, in the observational study, no one from the group without the

flyer bought a ticket. Based on this information, what should you do? Should you hand out flyers at ‘Zeus Theater’, or should you not? One intuition would be to just go with the experimental data. After all, RCTs are the gold standard of causal inference. Based on the experimental data, handing out flyers might seem worth it. More people with the flyer bought the ticket than did people without it. However, we should also consider the cost of handing out flyers and the potential consequences, such as damaging the theater’s reputation if the wrong person was handed a flyer. Can the data from the observational study help us make better decisions?

Before answering this question, let’s consider what could happen if we handed a flyer to a new customer. We will focus on situations where the action and outcome spaces are binary. You either hand out the flyer or you don’t, and the outcome is either positive or negative. Figure 1B illustrates the four possibilities. We could hand a flyer to someone like Mary: a person who goes to the movie because of the flyer, and who wouldn’t go if she didn’t get one. In line with previous work (see Angrist, Imbens, & Rubin, 1996), we will refer to people like Mary as *compliers*. We could hand a flyer to someone like Jerry (an *always-goer*): a person who goes to see the movie no matter what. Or we could hand the flyer to someone like Larry (a *never-goer*): a person who doesn’t go to see the movie no matter what. Finally, we could hand the flyer to someone like Carrie (a *defier*): a person who *doesn’t go* to the movie because of the flyer, and who would have gone otherwise. Carrie may seem like an unusual case, but there are many reasons why someone could be put off by a flyer. For example, maybe Carrie liked the title of the movie but the flyer’s description made her realize she wouldn’t enjoy it.

It’s clear that the more compliers (like Mary) and the fewer defiers (like Carrie) there are, the greater the likelihood that handing out flyers will be worthwhile. Also, given that handing out flyers is costly, the proportion of compliers has to be high enough overall – handing out flyers to always-goers and never-goers is a waste. With this in mind, we can now rephrase the earlier question: does having information from both experimental and observational studies allow us to better estimate the proportion of the four different types of people responding to flyers? It turns out, the answer is ‘yes’. If we correctly use the information from both types of studies, we will make better decisions. We will demonstrate how when we describe the CDM. But before doing so, we want to briefly discuss prior work on causality and counterfactuals in decision-making.

Causality and counterfactuals in decision-making

People’s causal beliefs are critical for decision-making. Here, we focus on how people learn about causal relationships, and how they use causal information for making decisions.

Learning about causal relationships

Work in psychology has studied how people make causal judgments based on contingency information like that shown in Figure 1A (P. W. Cheng, 1997; Griffiths & Tenenbaum, 2005; Jenkins & Ward, 1965). Several models of how people make these judgments have been developed. For example, the ΔP model (Jenkins & Ward, 1965; Shanks, 1986) predicts that people’s causal strength judgments track $\Delta P = p(y|x) - p(y|x')$, where x and x' represent the presence and absence of the cause, and y the presence of the effect. The power

PC model (P. W. Cheng, 1997) predicts that people track “causal power” – the probability that the cause produces the effect in the absence of other causes. For generative causes (as opposed to preventative ones), this quantity is defined as $\text{power} = \frac{\Delta P}{1 - p(y|x')}$. Griffiths and Tenenbaum (2005) developed a model of causal support that unifies both accounts. The model compares how much evidence there is for a causal graph with a generative link from cause to effect versus an alternative causal graph where that link is missing. Depending on what assumptions one makes about how the cause’s influence combines with other factors (additively or via a noisy-or function), the causal support model’s predictions align with those of ΔP or power.

Importantly, all of this work assumes that causal judgments are based on a single source of contingency information. Moreover, the models do not distinguish how the contingency information was generated: whether it was the result of an observational study, or a randomized controlled trial. This means that these models don’t have a principled mechanism for combining information from observational and experimental studies (beyond forming a single contingency table that combines the cases in each of the four cells across both studies; see Bareinboim & Pearl, 2016). While we focus here on covariation information (Lagnado, Waldmann, Hagmayer, & Sloman, 2007), people’s causal models are also informed by prior knowledge (Bramley, Gerstenberg, Tenenbaum, & Gureckis, 2018; Griffiths & Tenenbaum, 2009; Ullman, Stuhlmüller, Goodman, & Tenenbaum, 2018), temporal information (Bramley, Gerstenberg, Mayrhofer, & Lagnado, 2018; Gong, Gerstenberg, Mayrhofer, & Bramley, 2023; Hagmayer & Waldmann, 2002), mechanistic information (Ahn, Kalish, Medin, & Gelman, 1995), and different kinds of social information, such as explanations (Keil, 2006; Kirfel, Icard, & Gerstenberg, 2022) and demonstrations (Gweon, 2021).

Making decisions with causal information

Once people have formed beliefs about how a system works, how does this causal knowledge support their decision-making (Sloman & Hagmayer, 2006)? According to Pearl (2018), causal knowledge lives in a three-level hierarchy (see also Bareinboim, Correa, Ibeling, & Icard, 2022; Pearl & Mackenzie, 2018). An agent with knowledge on level I can form associations and make predictions. For example, they can compute the predictive relationship between x and y , assessing quantities like $p(y|x)$. On level II, the agent can compute the consequences of (hypothetical) interventions, such as the probability $p(y_x)$ that Y would take on value y given an intervention setting X to x . This means such an agent can anticipate what would happen if they were to perform a certain action. Finally, on level III, the agent can answer counterfactual questions, such as whether y would still have happened if x hadn’t happened, given that in fact, both x and y happened, notated as $p(y_{x'}|x, y)$.

The difference between level I and level II is captured in the adage that correlation is not causation. People intuitively distinguish between observational probabilities (level I) and interventional probabilities (level II). For example, when X and Y are correlated, there are three common reasons why this might be the case: X causes Y , Y causes X , or X and Y are the effects of a common cause C . People are much more likely to intervene on X (setting $X = x$ with the goal of making $Y = y$) when they believe that there is a causal link from X to Y compared to when X and Y are effects of a common cause (Hagmayer & Sloman, 2005).

The difference between level II and level III is more subtle. When it comes to making decisions, one may wonder how counterfactuals could be helpful. When we make decisions, we think about the future, whereas counterfactuals are generally evaluations of the past (Gerstenberg, 2022). In regret theory (Loomes & Sugden, 1982), an alternative framework to standard expected utility theory (Von Neumann & Morgenstern, 1947), decision-makers choose actions that minimize expected regret. Regret is a counterfactual notion. To regret having chosen x , a person must have had an alternative, x' , and believe that choosing that alternative would have resulted in a better outcome (Zeelenberg, Beattie, van der Pligt, & de Vries, 1996). Regret is not the only counterfactual concept that affects decision-making. In fact, in many instances, the decision outcome that an agent wants to maximize is inherently counterfactual (Bradley & Stefánsson, 2017). For example, consider an agent who wants to minimize harm (or maximize benefit). “Harm” here is not merely the negative utility of some outcome. It’s a counterfactual notion: a person is harmed by the agent’s choice if they would have been better off had the agent chosen otherwise (Beckers et al., 2023, 2024; Richens et al., 2022; Wu, Sridhar, & Gerstenberg, 2023).

Prior research shows that people are sensitive to whether a bad outcome arises through action or through inaction, even when this distinction has no bearing on expected outcomes. Ritov and Baron (1990) asked participants to consider a hypothetical disease that was expected to kill a certain number of children, and a vaccine that eliminated this risk but carried its own small risk of death as a side effect. Participants reported the maximum level of risk that should be tolerated by the government in order to institute a compulsory vaccination program. A decision-maker who simply wanted to minimize the expected number of deaths should be willing to vaccinate as long as the vaccine’s risk of death was lower than the disease’s risk of death. Instead, participants demanded that the vaccine be considerably safer than the disease before they were willing to vaccinate, and a substantial proportion refused to vaccinate even when the vaccine was much safer than the disease. Importantly, participants were much more likely to recommend giving the vaccine when they were informed that children who were susceptible to death from the disease were the same children susceptible to death from the side effects of the vaccine, compared to a situation where some children might die from the vaccine who wouldn’t have died otherwise. Ritov and Baron (1990) termed this asymmetry *omission bias* (see also Baron & Ritov, 1994, 2004; Ritov & Baron, 1992, 1995): people weigh harm that results from their own action (a child dying because of the vaccine) more heavily than equivalent, or even greater, harm that results from inaction (a child dying because they were not vaccinated).

Importantly, it’s only on the third level of Pearl’s (2018) hierarchy that we can make sense of these results. People don’t want to make a decision where the negative outcome happens *because* of what they did. Applied to the example of the different types of movie-goers we introduced earlier, the concepts of *complier*, *always-goer*, *never-goer*, and *defier* are counterfactual concepts. What it means to be a complier in the context of our example is to be someone who went to see the movie *because* they received the flyer and who wouldn’t have gone otherwise. In a similar vein, a patient who underwent a drug treatment may ask themselves whether they got better because of the treatment or whether their condition would have improved anyhow. As we will see now, data from experimental and observational studies provides critical information about such counterfactuals.

The counterfactual decision model (CDM)

Figure 1 outlines how the counterfactual decision model works. The model tackles the question of how one should optimally combine experimental and observational data to make a good decision about whether or not to act. When other people are the targets of our actions, we are typically interested in identifying compliers: individuals who respond positively if encouraged, and who would respond negatively otherwise. We also want to avoid giving the treatment to defiers: individuals who respond negatively to treatment, and who would respond positively otherwise. Of course, we can never know whether a person actually was a complier or a defier, as we don't get to observe the counterfactual outcome – the fundamental problem of causal inference is that we only ever get to observe the outcome of a single treatment for an individual. Nonetheless, Balke and Pearl (1994) showed how experimental and observational data can be used to compute bounds on the probability that a person is a *complier* (see also Tian & Pearl, 2000):

$$\max \left\{ \begin{array}{l} 0, \\ p(y_x) - p(y_{x'}), \\ p(y) - p(y_{x'}), \\ p(y_x) - p(y) \end{array} \right\} \leq p(\text{complier}) \leq \min \left\{ \begin{array}{l} p(y_x), \\ p(y_{x'}), \\ p(x, y) + p(x', y'), \\ p(y_x) - p(y_{x'}) + p(x, y') + p(x', y) \end{array} \right\}. \quad (1)$$

Notice that Equation 1 features both experimental probabilities (e.g., $p(y_x)$, highlighted in blue) and observational probabilities (e.g., $p(y)$, highlighted in green). If all we had was observational probabilities, then all the terms with the experimental probabilities would drop, and we would have $0 \leq p(\text{complier}) \leq p(x, y) + p(x', y')$. Similarly, if we only had experimental probabilities, the terms with the observational probabilities would drop.

Knowing the probability that someone is a complier and the experimental probability $p(y_x)$, we can compute the probability of an *always-goer*:

$$\begin{aligned} p(y_x) &= p(\text{complier}) + p(\text{always-goer}) \\ p(\text{always-goer}) &= p(y_x) - p(\text{complier}). \end{aligned} \quad (2)$$

The reasoning here is that there are only two ways in which a person in the experimental group ended up going to the movie when they received a flyer y_x : they must have either been a complier (who was encouraged by the flyer) or an always-goer (who would have gone anyway).

To compute the probability that someone is a *defier* we make use of the following relationship:

$$\begin{aligned} p(y_x) - p(y_{x'}) &= p(\text{complier}) + p(\text{always-goer}) - [p(\text{defier}) + p(\text{always-goer})] \\ p(y_x) - p(y_{x'}) &= p(\text{complier}) - p(\text{defier}) \\ p(\text{defier}) &= p(\text{complier}) - [p(y_x) - p(y_{x'})]. \end{aligned} \quad (3)$$

Here, we used Equation 2 to replace $p(y_x)$ and the knowledge that there are only two ways in which a person in the experimental group ended up going to the movie when they didn't receive a flyer $y_{x'}$: they were either a defier or an always-goer. The difference in the probability of a positive outcome between the two experimental groups $p(y_x) - p(y_{x'})$ is also called the average treatment effect (ATE).

Finally, we can compute the probability that someone is a *never-goer*:

$$\begin{aligned} p(y'_x) &= p(\text{defier}) + p(\text{never-goer}) \\ p(\text{never-goer}) &= p(y'_x) - p(\text{defier}). \end{aligned} \tag{4}$$

Here, the reasoning is that there are only two ways in which a person who received a flyer ended up *not* going to the movie y'_x : they were either a defier or a never-goer.

In order to combine experimental and observational probabilities like we did in Equation 1, we need to make a *consistency* assumption (Mueller & Pearl, 2023):

$$\begin{aligned} p(y|x) &= p(y_x|x) \\ p(y|x') &= p(y_{x'}|x'). \end{aligned} \tag{5}$$

This captures the intuition that the way in which a flyer was received shouldn't affect a person's probability of going to the theater. In other words, the outcome of a person in the observational group who chose the flyer would have been the same had they been given the flyer in the experiment. And, similarly, the outcome of a person who didn't choose the flyer in the observational group would have been the same had they not been given the flyer in the experiment. Further, we make the assumption that *each individual acts deterministically*. For example, a complier buys a ticket with probability 1 if they receive a flyer, and with probability 0 if they don't receive one.

Now that we know how to compute the probability of the different types of people based on experimental and observational data (see Figure 1B), how can we use this information to make good decisions? We need to assign a value to the four different cases (see Figure 1C). For example, in our movie-goer scenario, it would make sense to assign a positive value β to giving a flyer to a complier (because we gain a customer), a small negative value γ to an always-goer and θ to a never-goer (because we wasted the flyer on them), and a larger negative value δ to a defier (because we lost a customer). Based on these values, we can then compute the expected benefit of handing out the flyer, like so

$$\begin{aligned} \text{expected benefit} &= \beta \cdot p(\text{complier}) + \gamma \cdot p(\text{always-goer}) + \theta \cdot p(\text{never-goer}) + \delta \cdot p(\text{defier}) \\ &= \beta \cdot p(y_x, y'_{x'}) + \gamma \cdot p(y_x, y_{x'}) + \theta \cdot p(y'_{x'}, y'_{x'}) + \delta \cdot p(y'_{x'}, y_{x'}), \end{aligned} \tag{6}$$

where we've written out the different types of people following the notation in prior work (Angrist et al., 1996; Li & Pearl, 2022; Mueller & Pearl, 2023; Tian & Pearl, 2000). For example, the probability of being a complier, $p(y_x, y'_{x'})$, is the probability that a person would get a ticket (y) if they got a flyer (x) but wouldn't get a ticket (y') if they didn't get a flyer (x'). So, to make good decisions, we need to estimate the probability of the four different types of people, assign a value to each type, compute the expected benefit, and act accordingly.¹

While we generally benefit from having both experimental and observational data, there are some conditions where having observational data in addition to experimental data doesn't help. This is the case when either *monotonicity* (Angrist et al., 1996) or *gain equality*

¹Sometimes, we may not have point estimates of the probabilities of the different types of people based on the data, because the bounds are not tight. Li and Pearl (2022) show how to compute the benefit function in the more general case.

holds (Li & Pearl, 2019). Monotonicity means that the probability of being a defier is 0 (i.e., there are no movie-goers who were discouraged by the flyer – people who would have gone had they not received it). In other words, it’s not possible for a change from x' to x to lead to a change from y to y' . If $p(\text{defier}) = 0$, then this means that $p(y_x) - p(y_{x'}) = p(\text{complier})$ (see Equation 3), so all we need to know is the average treatment effect based on the experimental data alone. Gain equality holds when the benefit of selecting a complier and a defier is equivalent to the benefit of selecting an always-goer and a never-goer. In other words, gain equality is satisfied when $\beta + \delta = \gamma + \theta$ in Equation 6.

A worked example

To illustrate how the model works, let’s consider the data from ‘Zeus Theater’ in Figure 1A again. Our first goal is to determine the probability of the four counterfactual types of people who go to ‘Zeus Theater’. Let’s start with computing the probability that someone is a complier (see Equation 1). To do so, we need the experimental and observational data. The experimental study at ‘Zeus Theater’ establishes that $p(y_x) = 0.7$ and $p(y_{x'}) = 0.5$. If that were all we had, then we could use Equation 1 to determine that the probability of someone being a complier is within the following range: $\max\{0, 0.2\} \leq p(\text{complier}) \leq \min\{0.7, 0.5\}$. So the chance of someone being a complier is between 20% and 50%. Notice that to determine the bounds here, we simply dropped all of the terms that have observational probabilities from Equation 1.

In the observational data, the probability of a movie-goer taking a flyer of their own accord is $p(x) = 0.5$ as half of the movie-goers took a flyer and the other half didn’t. We can compute the observational probabilities we need for Equation 1 in the following way: $p(x, y) = p(y|x) \cdot p(x) = 0.4 \cdot 0.5 = 0.2$, $p(x', y) = 0$, $p(x, y') = 0.3$, $p(x', y') = 0.5$, and $p(y) = p(x, y) + p(x', y) = 0.2$. If all we had was this observational data, we would infer that $\max\{0\} \leq p(\text{complier}) \leq \min\{0.7\}$. So, based on the observational data only, we would infer that the probability of someone being a complier is between 0% and 70%.

Based on the information from the experimental data and observational data considered separately, we know already that the probability of someone being a complier must be between 20% and 50%. But, as it turns out, we can do better than that. By combining both experimental and observational data, we get the following: $\max\{0, 0.2, -0.3, 0.5\} \leq p(\text{complier}) \leq \min\{0.7, 0.5, 0.7, 0.5\}$. So, by combining both sources of the data, we get a point estimate of the probability that someone is a complier. Here, there is exactly a 50% chance that someone is a complier.

From $p(\text{complier}) = 0.5$ and $p(y_x) = 0.7$, we can compute $p(\text{always-goer}) = 0.2$ (using Equation 2), because there are only two ways a person with the flyer ends up going to the theater: they are either a complier or an always-goer. Knowing that the average treatment effect is $p(y_x) - p(y_{x'}) = 0.2$ and that $p(\text{complier}) = 0.5$, we can compute $p(\text{defier}) = 0.3$ (using Equation 3). Finally, from $p(\text{defier}) = 0.3$ and $p(y'_{x'}) = 0.3$, we can compute $p(\text{never-goer}) = 0$ (using Equation 4). Again, there are only two ways that someone with the flyer would *not* go to the theater: they must either be a defier or a never-goer. So, based on the observational and experimental data, there is a 50% chance that someone is a complier, a 20% chance that they are an always-goer, a 30% chance they are a defier, and a 0% chance a never-goer.

Now that we know the probability of the four different types, we need to assign a value for each of them to decide whether it's worth handing out the flyer (Figure 1C). Let's assume that the cost of handing out the flyer is \$0.50, and that the reward of giving the flyer to a complier is \$10. The cost of handing the flyer to a defier is \$20. Assume this is due to a lost profit of \$10 and a \$10 damage to the theater's reputation for recommending a movie that turned a customer away. So, taking into account the cost of the flyer, we get the following values for the different types of movie-goers shown in Figure 1C: Mary the complier ($\beta = \$9.50$), Jerry the always-goer ($\gamma = -\$0.50$), Larry the never-goer ($\theta = -\$0.50$), and Carrie the defier ($\delta = -\$20.50$). Putting the values and probabilities into Equation 6, we get: expected benefit = $\$9.50 \cdot 0.5 - \$0.50 \cdot 0.2 - \$0.50 \cdot 0 - \$20.50 \cdot 0.3 = -\$1.50$. So, it's not worth handing out flyers at 'Zeus Theater' (see Figure 1D). We expect to lose \$1.50 in value per customer by handing out flyers compared to not handing out flyers. This example illustrates that we can make better decisions by integrating both experimental and observational data, rather than relying on either data source alone. Now, we will examine how people make decisions, and whether they follow the CDM's normative predictions.

Experiment 1: Decisions with and without feedback

In this experiment, participants viewed situations like the ones in Figure 1A and had to decide whether or not to hand a flyer to a movie-goer from that theater. Between conditions, we manipulated whether participants received feedback about their choices. We explore to what extent participants' decisions are predicted by the CDM versus alternative models, and whether receiving feedback affects their performance.

Methods

All three experiments in this paper were pre-registered. You can access the pre-registrations, experiment demos, data, materials, statistical analyses and an interactive visualization of the CDM at https://github.com/cicl-stanford/good_decisions. All experiments were approved by Stanford's Institutional Review Board (IRB #48665).

Participants

We recruited 199 participants via Prolific (*age*: $M = 35$ years, $SD = 12$; *gender*: 116 female, 79 male, 2 non-binary, 2 undisclosed; *race*: 162 White, 13 Black, 12 Asian, 5 Hispanic, 2 American Indian/Alaska Native, 1 Native Hawaiian/Pacific Islander, 1 Multiracial, 3 undisclosed). Participants received a base payment of \$2.50 and an initial bonus of \$1. Whenever they handed out the flyer, their bonus changed by that trial's benefit in Table 1 divided by 10, and declining to hand it out left their bonus unchanged. For example, handing out the flyer on trial 1 would cost them 55 cents, and handing it out on trial 22 would earn them 35 cents. Making every decision correctly would bring their bonus to \$2.55, and making every decision incorrectly would in principle bring it to $-\$1.90$. However, the bonus could not fall below \$0.

Design

Participants were assigned to the 'feedback' condition ($N = 100$) or 'no feedback' condition ($N = 99$). Participants saw 22 trials where the information was presented just

like in Figure 1A. Each scenario took place at a different movie theater. In each scenario, participants saw experimental data (randomly chosen movie-goers were handed a flyer) and observational data (movie-goers chose whether to take a flyer). We randomized the trial order between participants. The trials varied the experimental and observational data (see Table 1).¹ Experimental data is represented by the number of movie-goers who returned to watch the movie, separately for those who were handed a flyer and those who weren't. Observational data is represented the same way, separately for those who took a flyer and those who didn't. We chose these scenarios with the following considerations in mind. We wanted a range of situations that differed in their expected benefit. We wanted to have the same number of trials with positive and negative expected benefit. And we wanted to have situations where the experimental probabilities were held constant, but the observational data differed in a way that affected whether the expected benefit was positive or negative.

Table 1

Experiment trial information and participant decisions. The trials are ordered based on the expected benefit (from low to high). $p(y_x)$ and $p(y_{x'})$ are the experimental probabilities; $p(y|x)$ and $p(y|x')$ the observational probabilities. x = cause present, and x' = cause absent; y = effect present, and y' = effect absent. $p(c)$ = probability of complier, a = always-goer, n = never-goer, d = defier according to the CDM. The last five columns show the proportion of participants who handed out the flyer in the different experimental conditions. Values in green indicate trials where the majority made the correct decision, and values in red where they were incorrect. The bottom row shows the proportion of correct choices together with the 95% bootstrapped confidence interval. In the 'Trial' column, the letters in round parentheses refer to the trials shown in Figure 2, and the ones in square brackets to the trials in Figure 6.

Trial	$p(y_x)$	$p(y_{x'})$	$p(y x)$	$p(y x')$	$p(c)$	$p(a)$	$p(n)$	$p(d)$	benefit	no feedback	feedback	data composition	people inference (novice)	people inference (expert)
1 [C]	0.5	0.5	0.0	0.0	0.5	0.0	0.0	0.5	-5.5	0.64	0.54	0.29	0.24	0.06
2 [A]	0.5	0.5	1.0	1.0	0.5	0.0	0.0	0.5	-5.5	0.20	0.30	0.17	0.46	0.09
3	0.5	0.4	0.0	0.0	0.5	0.0	0.1	0.4	-3.5	0.58	0.51	0.50	0.35	0.36
4	0.6	0.5	0.2	0.0	0.5	0.1	0.0	0.4	-3.5	0.66	0.60	0.52	0.46	0.47
5	0.6	0.5	1.0	1.0	0.5	0.1	0.0	0.4	-3.5	0.41	0.44	0.38	0.54	0.32
6	0.6	0.4	0.2	0.0	0.55	0.05	0.05	0.35	-2.0	0.73	0.68	0.73	0.46	0.57
7 [E]	0.7	0.5	0.4	0.0	0.5	0.2	0.0	0.3	-1.5	0.78	0.78	0.81	0.69	0.66
8 (A)	0.7	0.5	1.0	1.0	0.5	0.2	0.0	0.3	-1.5	0.37	0.53	0.40	0.63	0.53
9	0.5	0.3	0.0	0.0	0.5	0.0	0.2	0.3	-1.5	0.61	0.56	0.65	0.28	0.57
10 (D), [D]	0.5	0.5	0.0	1.0	0.0	0.5	0.5	0.0	-0.5	0.17	0.14	0.25	0.22	0.08
11 (E), [B]	0.5	0.5	1.0	0.0	0.0	0.5	0.5	0.0	-0.5	0.56	0.51	0.60	0.63	0.43
12	0.5	0.4	1.0	0.0	0.1	0.4	0.5	0.0	0.5	0.55	0.59	0.67	0.76	0.79
13	0.6	0.5	0.2	1.0	0.1	0.5	0.4	0.0	0.5	0.18	0.25	0.29	0.26	0.11
14	0.6	0.5	1.0	0.0	0.1	0.5	0.4	0.0	0.5	0.71	0.71	0.73	0.67	0.74
15	0.5	0.2	0.0	0.0	0.5	0.0	0.3	0.2	0.5	0.66	0.59	0.62	0.30	0.68
16	0.6	0.4	1.0	0.0	0.2	0.4	0.4	0.0	1.5	0.74	0.74	0.73	0.72	0.81
17 (C), [F]	0.7	0.5	0.4	1.0	0.2	0.5	0.3	0.0	1.5	0.28	0.41	0.29	0.35	0.34
18 (B)	0.7	0.5	1.0	0.0	0.2	0.5	0.3	0.0	1.5	0.79	0.71	0.79	0.81	0.85
19	0.5	0.3	1.0	0.0	0.2	0.3	0.5	0.0	1.5	0.62	0.62	0.77	0.72	0.81
20	0.7	0.3	1.0	0.6	0.6	0.1	0.1	0.2	1.5	0.61	0.68	0.65	0.80	0.85
21	0.5	0.2	1.0	0.0	0.3	0.2	0.5	0.0	2.5	0.66	0.65	0.75	0.63	0.87
22 (F)	0.7	0.3	1.0	0.0	0.4	0.3	0.3	0.0	3.5	0.77	0.77	0.88	0.80	0.91
Overall proportion correct										0.54 [0.51, 0.56]	0.55 [0.53, 0.57]	0.59 [0.55, 0.62]	0.58 [0.55, 0.62]	0.66 [0.64, 0.69]

¹For trials 6 and 20, the experimental and observational data don't yield a point estimate for the expected benefit. For trial 6, the bounds are $-\$2.50 \leq \text{benefit} \leq -\1.50 , and for trial 20 they are $\$0.50 \leq \text{benefit} \leq \2.50 . For these cases, we used the mean of the bounds as the expected benefit.

Procedure

All participants read a set of instructions about the task, saw an example trial, and answered comprehension questions. Participants were only allowed to proceed if they answered all comprehension questions correctly. If a participant answered one or more questions incorrectly, the instructions were presented again. Among other aspects of the task, the comprehension questions made sure that participants understood the difference between the experimental and the observational data (see Appendix A for the full set of questions).

Participants learned about a movie theater company that is testing out advertisements (flyers) for a new movie at various theater locations. Depending on the movie-goer, a flyer may encourage, discourage, or have no effect on whether the movie-goer will see the movie. Participants learned that a movie-goer who returns to see the advertised movie *because of the flyer* results in \$10 of profit. A movie-goer who doesn't return to see the advertised movie *because of the flyer* incurs a loss of \$20 due to lost profit and damaged reputation. The cost of a flyer is \$0.50. Hence, the associated benefits of compliers (β), always-goers (γ), never-goers (θ), and defiers (δ) are \$9.50, $-\$0.50$, $-\$0.50$, and $-\$20.50$, respectively. On each trial, participants' task was to decide whether or not to hand out a flyer to a new potential customer at the current theater. For example, we asked "Do you want to hand Marcus a flyer?", and participants responded by clicking a "yes" or "no" button. Participants were asked about a new individual at a new theater on each trial.

In the 'no feedback' condition, participants immediately proceeded to the next trial after having chosen their response. In the 'feedback' condition, participants learned whether their decision was correct, what their bonus in this round was, and what their bonus would have been had they chosen differently. For example, if a participant decided to hand out the flyer on trial 1, they received feedback that their choice was incorrect, that they lost 55 cents this round, and that they would have lost 0 cents if they hadn't handed out the flyer. If a participant decided *not* to hand out the flyer on trial 3, they received feedback that their choice was correct, that their bonus for this round was 0 cents, and that they would have lost 35 cents if they had handed out the flyer.

On the first two trials, participants viewed the data with a textual description first before making their decision. For example, participants were told that "0 out of 10 movie-goers who took the flyer went to see the movie, ..." (observational data) and that "5 out of 10 movie-goers who were given the flyer went to see the movie, ..." (experimental data). This was done to further ensure that participants understood how to read the data. The order of these first two trials was fixed (the first trial was trial 1, and the second trial was trial 10 from Table 1). After that, participants just viewed the data without any textual descriptions, and the trials were presented in random order.

At the end of the experiment, we asked what factors influenced their decisions and whether they had any other thoughts or comments about the experiment. Participants also provided demographic information. It took participants an average of 14 minutes ($SD = 7$) to complete the experiment.

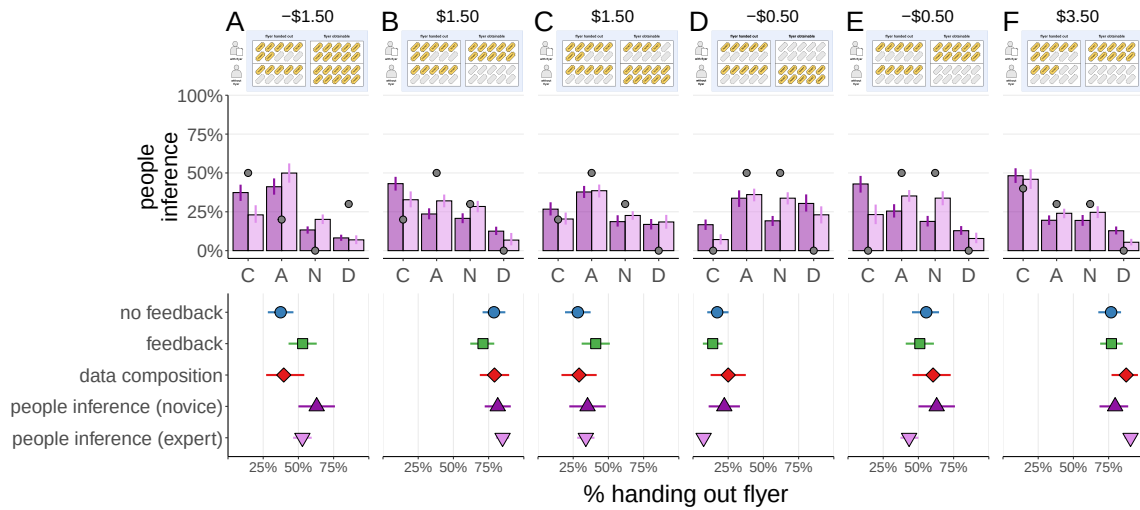


Figure 2

Experiment results for a selection of trials. The top row shows the information that participants saw about how many people returned to the theater to see the advertised movie based on whether or not they had been handed a flyer (experimental data) or had taken a flyer (observational data). The dollar value above shows the expected benefit of handing out a flyer in that theater. The bars in the second row show participants' average estimates from the 'people inference' condition of how likely the four different types of people are (purple bars = novices, pink bars = experts). C = complier, A = always-goer, N = never-goer, D = defier. Gray points show the CDM's predictions. The bottom row shows the probability with which participants decided to hand out the flyer in the different conditions. Note: Error bars show 95% bootstrapped confidence intervals.

Results

Figure 2 shows the results for a selection of six scenarios from the experiment. The top row shows the different scenarios, and the bottom row shows the percentage of participants who chose to hand out the flyer on each trial in the different experimental conditions. (You can ignore the middle row of the plot for now.) Scenarios A and B have the same experimental data but different observational data. In scenario A, the correct decision is *not* to hand out the flyer, as the expected benefit is $-\$1.50$. In the 'no feedback' condition, 37% of participants handed out the flyer, and 53% of participants did so in the 'feedback' condition. In scenario B, the correct decision is to hand out the flyer. The majority of participants in both conditions chose to do so with 79% in the 'no feedback' condition, and 71% in the 'feedback' condition. This comparison illustrates that participants weren't making their decisions solely based on the experimental data. That said, participants struggled to integrate the observational data as prescribed by the CDM. For example, in scenario C, the expected benefit is the same as in scenario B, even though the observational data is quite different. However, participants were much less likely to hand out the flyer in scenario C compared to scenario B.

As another illustration that participants struggled with optimally combining both

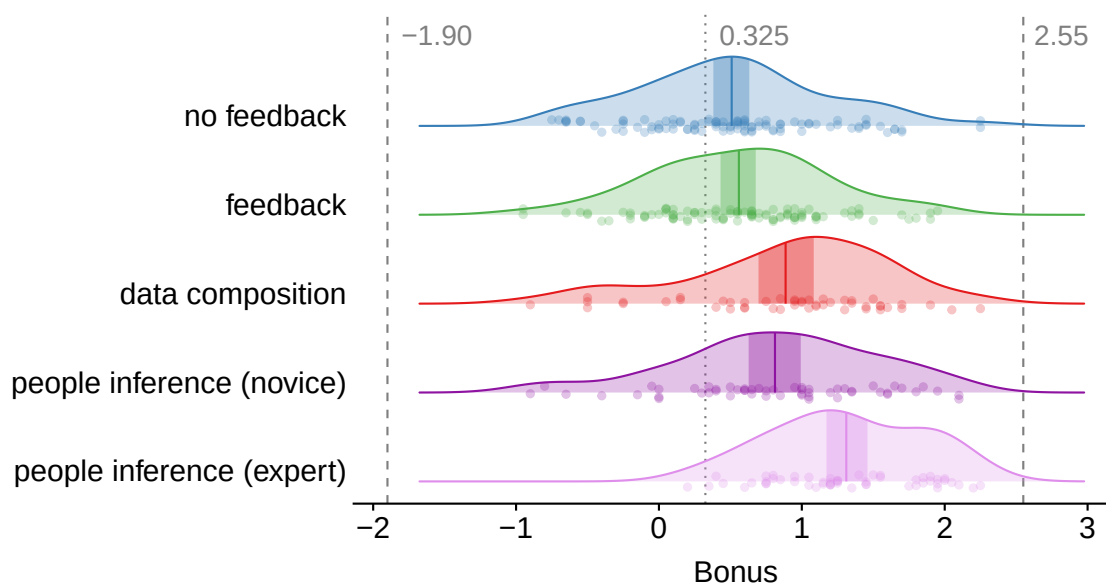


Figure 3

Distribution of final bonus scores. Small points show how much bonus in \$ each individual received. The solid vertical line shows the mean bonus for each condition. The ribbon around the mean marks the bootstrapped 95% confidence interval. The range of possible bonus amounts goes from $-\$1.90$ (always deciding incorrectly) to $\$2.55$ (always deciding correctly), with a random baseline of $\$0.325$. Note: The bonus is based on receiving $\$1$ as a start, and then adding or subtracting to that bonus based on the benefit of each decision as shown in Table 1 divided by 10. In the experiment, participants' minimum bonus was capped at $\$0$, so that they would not receive less money than their base payment.

sources of evidence, consider scenarios D and E. The expected benefit of both scenarios is the same ($-\$0.50$), so participants should not hand out the flyer. However, participants' decisions differ strongly between the two cases. In scenario D, 17% of participants in the 'no feedback' condition, and 14% in the 'feedback' condition handed out the flyer. In contrast, in scenario E, 56% of participants chose to do so in the 'no feedback' condition, and 51% in the 'feedback' condition. Participants' responses are intuitive here. Just like in scenarios A–C, the experimental data is identical in both scenarios D and E. The only thing that differs is the observational data. In scenario D, no one who took the flyer in the observational study bought a ticket, and everyone who didn't take the flyer bought a ticket. In scenario E, the observational data is completely flipped. Nonetheless, it turns out that the expected benefit is the same in both situations. The combination of both data sources implies that half of the people at each theater are always-goers, and the other half are never-goers. So we would be wasting our flyer in both scenarios. The difference in observational data is due to the always-goers not picking up a flyer (there's no need) and never-goers picking up a flyer (maybe they're hoping to have their minds changed) in scenario D, and the always-goers picking up a flyer (maybe they're excited about the movie) and never-goers not picking up a flyer (there's no need) in scenario E. Scenario F shows a case in which the majority of participants made the correct decision. In each condition, 77% of participants handed out

the flyer. This is the scenario with the highest expected benefit of \$3.50 (see Table 1).

Figure 3 shows how much bonus participants received in each condition. The expected bonus for random responding is \$0.325. Participants’ average bonus exceeded the chance level in both the ‘no feedback’ condition (Mean = 0.51, 95% confidence interval = [0.37, 0.64]) and the ‘feedback’ condition (0.55, [0.44, 0.68]). There was no credible difference between how much bonus participants received in the two conditions (see Table C1). In the ‘no feedback’ condition, 62% of participants received a bonus above the chance level, and 64% in the ‘feedback’ condition. Participants’ decisions in both conditions were highly correlated, $r = .94$.

Overall, participants in the ‘no feedback’ condition made the correct decision 54% [51%, 56%] of the time. Participants in the ‘feedback’ condition chose correctly 55% [53%, 57%] of the time. Participants in the ‘feedback’ condition did not make better decisions over time. We fitted a Bayesian logistic mixed effects model with trial number as fixed effect as well as random slopes and intercepts.¹ Participants’ likelihood of making the correct decision did not change as a function of trial number, 0.00 [−0.01, 0.01]. Participants were more likely to hand out the flyer than not, both in the ‘no feedback’ condition (56%, [53%, 58%]) and in the ‘feedback’ condition (56%, [54%, 58%]). Remember that we had chosen the trials so that, for half of them, it was the correct decision *not* to hand out a flyer (trials 1–11 in Table 1), and to hand it out for the other half (trials 12–22).

To better understand how participants made their decisions, we compared their decisions to the predictions of five different models (see Table 2). The CDM computes the expected benefit by combining the experimental and observational data in a non-linear way as described above. The other models use linear combinations of the experimental and observational data to predict participants’ decisions. Figure 4 shows scatterplots of model predictions and participants’ choices across the 22 trials. Because the models differ in their

Table 2

Models of participants’ decisions. *Each model was fitted as a Bayesian mixed effects logistic regression which included a global intercept and random intercepts for participants. The models differed in their fixed effects. For example, the ‘Experimental’ model has two predictors and we fitted the weight on each predictor to the data. We indicate separate predictors here by using a + sign as is customary in linear regression models. For the ‘ ΔP combined’ model, we first computed ΔP separately for the experimental data and the observational data (e.g., $\Delta P_{\text{exp}} = p(y_x) - p(y_{x'})$), and then used each quantity as a predictor in the model.*

Name	Predictors	Description
Experimental	$p(y_x) + p(y_{x'})$	Only uses experimental data.
Observational	$p(y x) + p(y x')$	Only uses observational data.
Full	$p(y x) + p(y x') + p(y_x) + p(y_{x'})$	Uses both experimental and observational data.
ΔP combined	$\Delta P_{\text{exp}} + \Delta P_{\text{obs}}$	Computes ΔP for both experimental and observational data.
CDM	expected benefit	Computes the normative benefit function (see Equation 6).

¹All Bayesian models were written in Stan (Carpenter et al., 2017) and accessed with the **brms** package (Bürkner, 2017) in R (R Core Team, 2019). We refer to an effect as ‘credible’ when the 95% credible interval of the corresponding posterior distribution excludes 0.

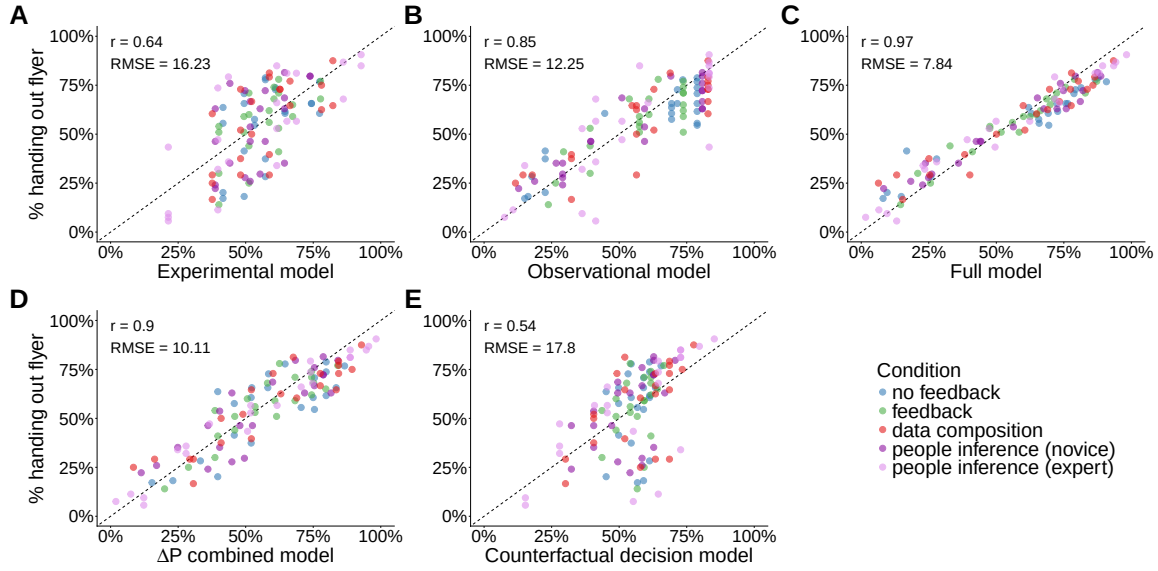


Figure 4

Relationship between model predictions and participants' decisions. Each point shows the proportion of participants who handed out a flyer for a given trial, separately for the different experimental conditions. Model predictions are based on Bayesian mixed effects logistic regressions (see Table 2).

number of free parameters, we compared their performance using approximate leave-one-out cross-validation. Table 3 summarizes how well the models capture participants' choices. The results show that in both the 'no feedback' and the 'feedback' condition, the 'full' model best explains participants' decisions. The CDM fares the least well. It correlates poorly with participants' decisions in both conditions.¹

Discussion

Participants had trouble using both experimental and observational data to make good decisions. While they performed better than chance on average, performance was far from optimal. Interestingly, participants' performance wasn't much better when they received feedback, and didn't improve over time in the feedback condition. Integrating the two sources of data as prescribed by the CDM is not something that comes naturally to people. In the words of one of our participants: "It doesn't feel consistent. There were two sets of almost identical data; I chose one answer, got it wrong, then choose the other and got it wrong again." Even though participants' decisions didn't follow the CDM, their answers were very systematic. Some participants explicitly mentioned focusing on one side of the data, observational (e.g., "I looked at how many [movie-goers] came back if flyers were available.") or experimental (e.g., "I looked at how many [movie-goers] who returned

¹Even if all five free parameters in the 'full' model are fitted, it still only correlates $r = .71$ with the predictions of the CDM.

were given fliers.”). Overall, the ‘full’ model, which combines the information from both experimental and observational studies in a weighted additive way, captured participants’ decisions best (see Table 3).

Experiment 2: Inferring counterfactual types before making decisions

In Experiment 1, we found that people had trouble integrating experimental and observational data to make optimal decisions. In Experiment 2, we wanted to see whether people can estimate the probability of the different counterfactual types (complier, always-goer, etc.) based on the data, and whether being asked to do so would improve performance. We also asked another group of participants to report the data itself before making decisions, to see whether any potential improvement was due specifically to thinking about the counterfactual types, or merely due to being asked to pay closer attention to the data.

Methods

Participants

We recruited 102 participants via Prolific (*age*: $M = 36$ years, $SD = 12$; *gender*: 56 female, 42 male, 2 non-binary, and 2 undisclosed; *race*: 84 White, 6 Black, 10 Asian, 1 Hispanic, and 1 undisclosed). Participants were compensated at a base rate of \$10 per hour, with a maximum rate of \$17.10 per hour depending on performance. Of the 102 participants, 48 were assigned to the ‘data composition’ condition and 54 to the ‘people inference’ condition.

Design and procedure

The design and procedure of Experiment 2 were largely identical to those of the ‘feedback’ condition in Experiment 1. However, this time, participants in the ‘people inference’ condition were asked to assess the composition of the four types of people in each theater prior to choosing whether or not to hand out a flyer. On each trial, participants were asked to estimate percentages of people at this movie theater and enter their estimates in text boxes next to the four labels “encouraged to go by the flyer”, “would go with or without the flyer”, “would not go with or without the flyer”, and “discouraged to go by the

Table 3

Model comparison for each experimental condition. r = Pearson correlation, $RMSE$ = root mean squared error (computed on the trial averages), CDM = counterfactual decision model. $\Delta elpd$ shows the difference in expected log predictive density using approximate leave-one-out cross-validation between the best-fitting model (indicated by 0) and the other models, along with associated standard error. Larger negative numbers indicate worse performance (see Vehtari, Gelman, & Gabry, 2017, for details).

Model	no feedback			feedback			data composition			people inference (novice)			people inference (expert)		
	r	RMSE	$\Delta elpd$	r	RMSE	$\Delta elpd$	r	RMSE	$\Delta elpd$	r	RMSE	$\Delta elpd$	r	RMSE	$\Delta elpd$
Experimental	.49	0.17	−355 (23.3)	.62	0.13	−179.7 (17.8)	.64	0.16	−122.5 (14.8)	.51	0.17	−157.6 (15.7)	.77	0.18	−115.2 (14.3)
Observational	.90	0.12	−53.4 (10.5)	.84	0.10	−44.7 (9.8)	.86	0.12	−23.1 (7.4)	.92	0.10	−12.2 (5.4)	.80	0.17	−94 (13.2)
Full	.97	0.10	0 (0)	.97	0.06	0 (0)	.96	0.09	0 (0)	.98	0.07	0 (0)	.98	0.07	0 (0)
ΔP combined	.79	0.13	−220.8 (18.7)	.88	0.08	−92.2 (13.2)	.91	0.10	−21.6 (7.8)	.87	0.11	−58.1 (10.1)	.98	0.06	−0.1 (5.1)
CDM	.25	0.19	−394.3 (24.1)	.35	0.15	−213.4 (18.6)	.58	0.17	−124.6 (15.9)	.50	0.17	−154.1 (15.7)	.69	0.20	−142.5 (15.5)

flyer”. The input boxes were limited to numbers from 0 to 100, and participants could only proceed if the numbers they entered into the four boxes added up to 100.

In the ‘data composition’ condition, participants were asked to enter the number of people who returned to the theater under each condition. Participants put in a number between 0 and 10 next to the four labels “took flyer (top right cell)”, “did not take flyer (bottom right cell)”, “were given flyer (top left cell)”, and “were not given flyer (bottom left cell)”.

On the same screen, participants in both conditions were also asked to choose whether or not to hand out a flyer. In the ‘data composition’ condition, participants were only able to proceed to the next screen if they had entered all of the four data values correctly (and after having decided whether or not to hand out a flyer). After their decision, participants received feedback in the same way as in Experiment 1’s ‘feedback’ condition.

We made one small change in the order in which the trials were presented. Because all participants received feedback now, we wanted it to be the case that among the first two trials, the correct decision was to hand out the flyer in one of them, and not in the other. So participants always saw trial 12 first, and then trial 10 second (see Table 1). The order for the rest of the trials was randomized. Participants took an average of 29 minutes (SD = 14) to complete the experiment.

Results

We will first discuss participants’ inferences in the ‘people inference’ condition, then their decisions in both conditions of the experiment, and finally the relationship between people inferences and decisions.

People inference task

The middle row of Figure 2 shows participants’ inferences for a selection of trials together with the predictions of the CDM (Table B1 shows the inferences for all trials).

Table 4

***Predictors of participants’ people inferences.** Estimates of separate Bayesian linear mixed effects models, predicting participants’ inferences that a person is a complier, always-goer, never-goer, or defier, based on experimental and observational data. We standardized the predictors for this analysis. The values show the posterior means of the coefficients together with the 95% credible intervals. We bolded the strongest predictor in each model. r and RMSE for each model were computed on the level of the trial means.*

person	intercept	$p(y_x)$	$p(y_{x'})$	$p(y x)$	$p(y x')$	r	RMSE
Experiment 2 (Novices)							
complier	33.27 [30.89, 35.88]	1.81 [0.88, 2.79]	-0.62 [-1.61, 0.39]	8.91 [8.00, 9.84]	-2.90 [-3.91, -1.92]	.98	2.04
always-goer	27.34 [25.03, 29.56]	-0.63 [-1.54, 0.29]	2.32 [1.37, 3.28]	1.21 [0.33, 2.08]	6.74 [5.81, 7.63]	.98	1.74
never-goer	23.99 [21.87, 26.15]	-0.20 [-1.08, 0.67]	-1.79 [-2.64, -0.94]	-5.89 [-6.69, -5.07]	-4.18 [-5.05, -3.29]	.95	2.45
defier	15.21 [13.41, 17.00]	-1.00 [-1.70, -0.30]	0.09 [-0.61, 0.80]	-4.22 [-4.91, -3.52]	0.34 [-0.38, 1.05]	.86	2.67
Experiment 3 (Experts)							
complier	24.92 [22.30, 27.42]	4.43 [3.40, 5.48]	-5.11 [-6.18, -4.06]	5.02 [3.97, 6.04]	-3.72 [-4.83, -2.64]	.94	3.58
always-goer	32.02 [29.79, 34.31]	-0.95 [-1.89, -0.04]	4.82 [3.81, 5.81]	3.40 [2.47, 4.33]	5.30 [4.33, 6.27]	.93	3.57
never-goer	33.07 [30.51, 35.53]	-3.10 [-4.06, -2.15]	0.30 [-0.70, 1.31]	-5.18 [-6.12, -4.22]	-4.58 [-5.55, -3.61]	.97	2.24
defier	9.38 [7.04, 11.84]	-0.37 [-1.10, 0.40]	0.00 [-0.76, 0.74]	-3.25 [-3.94, -2.56]	3.00 [2.24, 3.76]	.83	2.90

For example, in scenario A, the CDM predicts that there is a 50% chance that a person is a complier, a 20% chance they are an always-goer, a 0% chance a never-goer, and a 30% chance a defier. Participants' averaged judgments differed from these predictions. They considered it roughly equally likely that a person was a complier or an always-goer, and unlikely that they were a never-goer or a defier.

For scenarios B and C, the CDM predicts the same inference. However, people's inferences differed markedly between the two scenarios. They were more likely to believe that a person is a complier than an always-goer in scenario B, but the opposite is true for scenario C. In scenarios D and E, the CDM again predicts the same inference, but people's inferences differed. In scenario D, participants considered it likely that the person could be a defier, whereas the CDM considers that impossible. In scenario F, people's inferences are more closely aligned with the predictions of the CDM.

Table 4 shows what aspects of the data most strongly affected participants' inferences. We ran separate Bayesian linear mixed effects models for each person type (complier, always-goer, never-goer, and defier), using the experimental and observational data as predictors. We find that participants' judgments of whether someone is a complier are most strongly affected by the proportion of people who bought a ticket after having taken the flyer in the observational study, $p(y|x)$. This is also true for people's judgments about never-goers and defiers, although here the relationship is negative. For always-goers, participants' judgments are most strongly affected by the proportion of people who bought a ticket even though they didn't pick up a flyer, $p(y|x')$.¹

So while participants paid attention to the different sources of information, they weren't able to reliably infer the probability of the four counterfactual types across the trials. The correlation between the CDM's predictions and participants' inferences was $r = .11 [-.10, .31]$.

Decision task

Figure 2 shows participants' decisions for a selection of trials. For most of these trials, the majority of participants made the same decisions as in Experiment 1. The correlation between participants' decisions in the two Experiment 2 conditions was $r(\text{people inference, data composition}) = .69$ (see Table D1).

Figure 3 shows that participants performed slightly better in Experiment 2 than in Experiment 1. In the 'people inference' condition, participants' average bonus was 0.81 [0.62, 0.99], and in the 'data composition' condition, it was 0.89 [0.69, 1.08]. In each of the 'people inference' and 'data composition' conditions, 81% of participants received a bonus above the chance level. Participants received credibly more bonus in the 'people inference' and 'data composition' conditions than in both Experiment 1 conditions (see Table C1). However, there was no credible difference in performance between the 'people inference' and the 'data composition' condition. Overall, participants in the 'people inference' condition decided correctly 58% [55%, 62%] of the time, and 59% [55%, 62%] of the time in the 'data composition' condition. Just like in Experiment 1, participants were more likely to

¹Notice that fitting a linear model based on the experimental and observational data to predict the true probability of the counterfactual types according to the CDM yields only moderate correlations, illustrating that these inferences cannot simply be achieved through a linear combination of features: $r(\text{complier}) = .47$, $r(\text{always-goer}) = .50$, $r(\text{never-goer}) = .50$, $r(\text{defier}) = .42$.

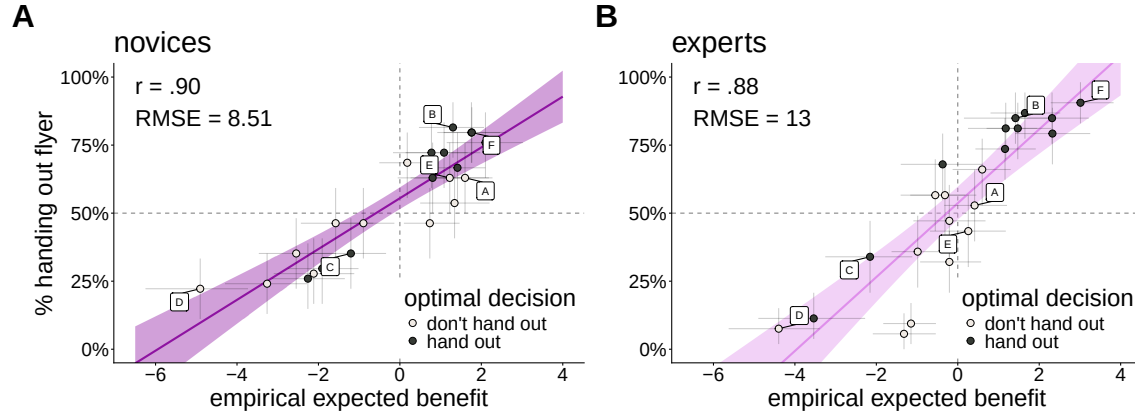


Figure 5

Relationship between people inference and decisions. Scatterplots show the relationship between the empirical expected benefit computed from participants' people inferences, and the probability with which participants handed out the flyer. **A** shows the results for novices and **B** for experts. Each point shows one of the 22 trials. Note: The error bars show 95% bootstrapped confidence intervals. The labels refer to the scenarios shown in Figure 2.

hand out the flyer than not, doing so 54% [51%, 56%] of the time in the 'people inference' condition, and 57% [54%, 59%] of the time in the 'data composition' condition. Just like in the 'feedback' condition in Experiment 1, participants' decisions did not improve across trials despite receiving feedback. There was neither an effect of trial number on whether a decision was correct in the 'people inference' condition, $-0.02 [-0.03, 0.00]$, nor in the 'data composition' condition, $-0.01 [-0.03, 0.01]$.

When comparing participants' decisions to the predictions of the CDM and alternative models (Table 3), we see again that the 'full' model captured participants' decisions best while the CDM performed much less well (see also Figure 4).

Relationship between people inferences and decisions

We saw that neither participants' people inferences nor their decisions are well-predicted by the CDM. However, participants' (incorrect) people inferences were closely related to their decisions. Figure 5 shows a very close relationship between the empirical expected benefit (computed from participants' inferences) and their decisions. To compute the empirical expected benefit, we took the averaged judgments for the four person types and multiplied them by the benefit for each type, as prescribed by Equation 6. For example, in scenario A of Figure 2, participants' averaged inferences were the following: $p(\text{complier}) = 0.37$, $p(\text{always-goer}) = 0.41$, $p(\text{never-goer}) = 0.13$, $p(\text{defier}) = 0.08$, so the empirical expected benefit for this trial would be: $\$9.50 \cdot 0.37 - \$0.50 \cdot 0.41 - \$0.50 \cdot 0.13 - \$20.50 \cdot 0.08 = \$1.60$. Notice that the actual expected benefit for this scenario, according to the CDM, is $-\$1.50$. The majority of participants decided to hand out flyers, which is in line with the empirical expected benefit but not the actual expected benefit.

Discussion

Having participants engage more closely with the experimental and observational data led to slightly improved decisions compared to those of Experiment 1. Even though participants did better in this experiment, their decisions were still only moderately correlated with the CDM’s predictions (see Table 3). While performance was higher this time compared to Experiment 1, it didn’t improve over time, suggesting that participants were unable to improve their decision-making strategy with feedback.

In the ‘people inference’ condition, we found that participants’ inferences were not well-predicted by the CDM. Instead, participants’ inferences were well-captured by a model that assumes a linear weighting of the observational and experimental probabilities (see Table 4). Furthermore, there was a highly systematic relationship between participants’ inferences and their decisions. The greater the expected benefit based on their people inferences, the more likely they were to hand out the flyer (see Figure 5).

Experiment 3: Expert judgments

Experiments 1 and 2 tested non-expert participants who were recruited via Prolific – an online crowdsourcing platform. We found that the counterfactual inferences and decisions of these participants weren’t well-captured by the CDM. In this experiment, we tested expert participants. Most people rarely make decisions based on covariation data from observational and experimental studies. However, data scientists, statisticians, doctors, and consultants have ample experience with this kind of data and regularly make high-stakes decisions based on it. Therefore, it is important to know whether their inferences and decisions correspond more closely to the predictions of the CDM. We recruited students and postdocs who study statistics, economics, or data science at Stanford University and gave them the same task as participants in the ‘people inference’ condition in Experiment 2. We hypothesized that experts’ inferences about the different counterfactual types would be more accurate than those of novices, and that experts would make better decisions, too.

Methods

Participants

We recruited 53 participants by advertising the study on department email lists (*age*: $M = 23$ years, $SD = 4$; *gender*: 26 female, 26 male, 1 non-binary; *race*: 26 White, 21 Asian, 3 Black/African American, and 3 Other; *ethnicity*: 3 Hispanic, 50 Non-Hispanic; *degree*: 27 undergraduate students, 1 Master’s student, 22 PhD students, 3 postdocs). Participants received an Amazon gift voucher worth \$20 as base payment plus a performance-related bonus of up to \$10.20.

Design and procedure

The design and procedure were identical to the ‘people inference’ condition in Experiment 2. This time, we collected more demographic data at the end of the study, including

information about their degree, their department, their area of study, and their familiarity with causal inference.¹

Results

Figure 2 shows participants’ inferences and decisions for a subset of trials. Overall, experts’ inferences and decisions were similar to those of novices. However, in scenario A, experts misjudged the proportion of every counterfactual type by more than novices did, yet they were somewhat less likely to hand out the flyer, which was the correct decision in this scenario. In scenario E, the experts were more accurate in their inferences and less likely not to hand out the flyer, which was the correct decision.

People inferences

Contrary to what we predicted, experts were not better than novices at inferring the counterfactual types. For each participant and trial, we computed the squared error between their people inference and the CDM’s predictions. We then looked at whether the errors were smaller for experts compared to novices. In fact, experts had slightly higher errors in their inferences compared to novices, although this difference was not credible (100.60, 95% CrI $[-2.10, 203.69]$). Across all of the 22 trials, the novice and expert inferences were strongly correlated ($r = .75$, RMSE = 0.08), but expert inferences were less correlated with the normative inferences compared to novices (novices: $r = .11$, RMSE = 0.22; experts: $r = .06$, RMSE = 0.23).

Table 4 shows what information experts relied on when making their inferences. There were a few notable differences between experts and novices. First, when inferring the probability of compliers, experts put more weight on the experimental data for the group that didn’t get the flyer, $p(y_{x'})$. Second, when inferring the probability of never-goers, they put more weight on the experimental data for the group that got the flyer, $p(y_x)$. Finally, when inferring the probability of defiers, they put more weight on the observational data for the group that didn’t take the flyer, $p(y|x')$.

Decisions

As predicted, experts made better decisions than novices. Table 1 shows that experts were more often correct than novices, and Figure 3 shows that they earned more bonus based on their decisions (see also Table C1, which shows that there was a credible difference in performance). Figure 4 shows how well the different models capture participants’ decisions. Interestingly, whereas in all previous conditions, the ‘full’ model captured participants’ decisions best, experts’ decisions were captured almost as well by the ‘ ΔP combined’ model (see Table 3). The correlation between expert and novice decisions was $r = .79$. Figure 5 shows that, just like for novices, experts’ decisions are predicted by their inferences. However, the relationship between inferences and decisions is slightly weaker in experts compared to novices.

¹The first phase of the first trial (trial 12 in Table 1), where participants see the data with a textual description, had an incorrect image. The correct image was shown in the second phase where participants provided their responses. As the data in Table 1 shows, this programming mistake seems not to have affected the expert participants’ performance.

Finally, we explored whether there were any differences in performance in our expert sample based on their degree (Table E1), or their self-rated familiarity with causal inference (Table E2). We found no credible differences.

Discussion

Expert participants recruited from Stanford in the economics, statistics, and data science departments made better decisions than participants recruited through a crowdsourcing platform online. However, interestingly, experts were slightly less accurate than novices in their inferences about the counterfactual types of people in the population. And, while their decisions were credibly better than those of novices, experts were still far from optimal. There were no credible differences in expert performance based on their degree (e.g., undergraduate vs. postdoc), or based on their self-reported familiarity with causal inference. Experts' decisions were very well-explained by the ' ΔP combined' model. This model first computes the differences in the probability of a positive outcome with and without the cause separately for the experimental group and the observational group, and then uses a linearly weighted combination of these difference scores to determine how to act.

General discussion

How do people make decisions based on experimental and observational data? We explored this question in three experiments, focusing on cases where these data logically imply a unique best action. Participants decided whether or not to hand out a flyer to a movie-goer at a theater. To decide well, participants needed to work out how likely the movie-goer was to be a complier (someone who goes because of the flyer), an always-goer, a never-goer, or a defier (someone who doesn't go because of the flyer), and then weigh each of these possibilities by what it was worth – the calculation prescribed by the counterfactual decision model (CDM; see Figure 1).

Participants' decisions were better than chance, but far from optimal. Receiving trial-by-trial feedback about the consequences of one's choice – including what would have happened had one chosen differently – didn't improve performance. Having participants engage more closely with the data before deciding helped, but only modestly. In Experiment 2, participants who first reported the composition of the data and those who first estimated the proportions of the four counterfactual types of movie-goers both earned credibly more bonus than participants in Experiment 1 (see Table C1), yet their accuracy still hovered just under 60%. Notably, simply having participants report the numbers in the table worked as well as having them reason explicitly about counterfactual types.

One possibility for the low performance is that participants simply ignored the observational data, as many people learn that experiments are the gold standard in science. If that had been the case, then their decisions would have been constant across scenarios that share the same experimental data, but they clearly weren't (see Figure 2). So while participants used both sources of information, they didn't follow the CDM's predictions. For example, scenarios B and C in Figure 2 imply the same expected benefit despite very different observational data, and participants were far more willing to hand out the flyer in B than in C. Scenarios D and E make the point even more sharply: the observational data is exactly reversed, the implied expected benefit is identical, and participants' decisions

differed by roughly 40 percentage points. Across all conditions, participants' choices were best captured by weighted-additive models, and much less well by the CDM (see Table 3).

Experiment 2 shows that participants fail at the inference step (Figure 1B) rather than at the valuation step (Figure 1C). Participants' estimates of the proportions of compliers, always-goers, never-goers, and defiers bore almost no relationship to the CDM's normative predictions, $r = .11 [-.10, .31]$. Their inferences were not arbitrary though – they varied systematically with the data (see Table 4) – they were just wrong. Participants' inferences of the different counterfactual types were well-explained by models that combine the observational and experimental data in a linearly additive manner. Participants' decisions followed coherently from these incorrect inferences: the expected benefit computed from participants' estimates of the four types predicted their choices very well (see Figure 5). So, participants are doing the right thing with the wrong estimates.

Expertise didn't change this picture in the way we expected. The statistics, economics, and data science students in Experiment 3 made credibly better decisions than novices, earning more bonus. But their inferences about the four counterfactual types were no more accurate than those of novices – if anything slightly less so. Also, performance was unrelated to degree or to self-reported familiarity with causal inference. What distinguished experts was the heuristic they brought to bear: their decisions were well-explained by the ' ΔP combined' model, which computes a difference score separately within the experimental and the observational data and then combines the two. Experts, in other words, seem to have arrived at a better rule of thumb rather than at a better grasp of the counterfactual quantities.

Developing intuition

Let's try to develop a better intuition for how to correctly combine observational and experimental data to infer the different counterfactual types that make up the population. While Equation 1 provides the normative solution for inferring the probability that a person is a complier, this equation isn't particularly intuitive. Let us first summarize some of the key relationships here again:

$$p(y_x) - p(y_{x'}) = p(\text{complier}) - p(\text{defier}) \quad (7)$$

$$p(y_x) = p(\text{complier}) + p(\text{always-goer}) \quad (8)$$

$$p(y_{x'}) = p(\text{defier}) + p(\text{never-goer}). \quad (9)$$

Notice that these equations just use experimental data, and we've seen them all before (they are the same as Equations 2–4). Now the key to using them well is to identify which of the four counterfactual types the experimental and observational data constrain most directly. Once we have a good sense for the probability of one of the counterfactual types, we can use the relationships above to infer the rest.

Let's consider the data from 'Apollo Theater' (Figure 6A). Here, the observational data provides a very strong constraint. In fact, it rules out never-goers, since everyone went. Given that we know that $p(\text{never-goer}) = 0$, we can infer that $p(\text{defier}) = 0.5$ (using Equation 9), then that $p(\text{complier}) = 0.5$ (using Equation 7; because $p(y_x) - p(y_{x'}) = 0$), and finally that $p(\text{always-goer}) = 0$ (using Equation 8). The observational data from 'Bacchus Theater' (Figure 6B) rules out defiers, because everyone with the flyer went,

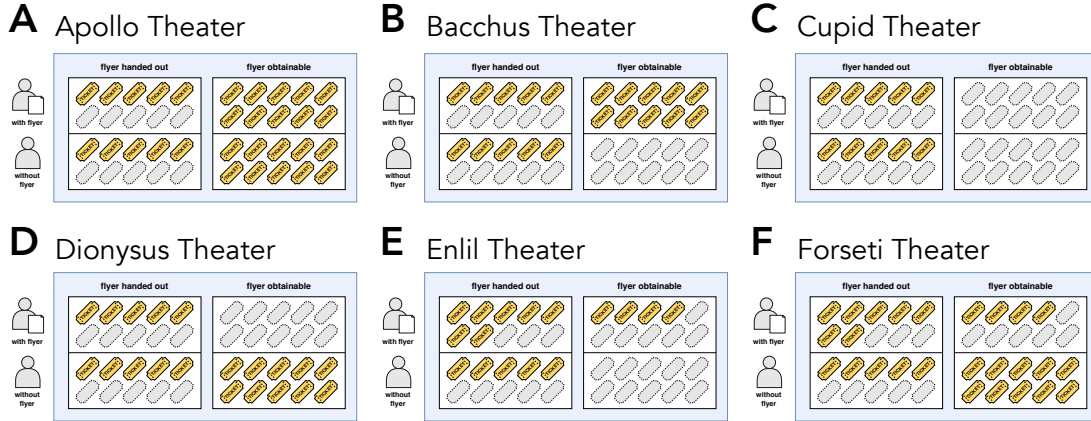


Figure 6

Observational and experimental data from six theaters. Some examples to build intuition about how experimental and observational data supports inferences about counterfactual types.

and no one without the flyer went. Given that we know that $p(\text{defier}) = 0$, we can infer that $p(\text{never-goer}) = 0.5$ (using Equation 9), that $p(\text{complier}) = 0$ (using Equation 7), and that $p(\text{always-goer}) = 0.5$ (using Equation 8).

The observational data from ‘Cupid Theater’ (Figure 6C) rules out always-goers (no one went). Notice that this means that, somewhat surprisingly, the inferences about the different counterfactual types are the same at ‘Cupid Theater’ as they are at ‘Apollo Theater’, even though the observational data looks very different. Whereas at ‘Apollo Theater’, all the compliers took the flyer, and all the defiers didn’t take the flyer, the opposite happened at ‘Cupid Theater’. Here, the defiers took the flyer (so they didn’t end up going), and the compliers didn’t take the flyer (so they also didn’t end up going). This situation might strike us as more unusual: why would the people whom a flyer dissuades take them, and those it persuades don’t? One explanation is that the movie’s title and flyer suggested different kinds of movies. The compliers found the title unappealing and skipped the flyer, even though it would have won them over, whereas the defiers were intrigued by the title, took a flyer, and were put off by its description. The counterfactual inferences at ‘Dionysus Theater’ (Figure 6D) and ‘Bacchus Theater’ are also the same. The data at ‘Dionysus Theater’ rules out compliers (no one with the flyer went, and everyone without the flyer went). At ‘Bacchus Theater’ all the always-goers took the flyer and the never-goers didn’t take the flyer, whereas at ‘Dionysus Theater’, the always-goers didn’t take the flyer and the never-goers took it.

Thanks to the logic of counterfactual inference, these four theaters were fairly easy to deal with. Things get a little more challenging when neither the observational nor the experimental data alone rule out one of the counterfactual types. Consider the data from ‘Enlil Theater’ (Figure 6E) and from ‘Forseti Theater’ (Figure 6F). The majority of participants made incorrect choices for both of these trials in all of the experiments. Notice that both feature the same experimental data but have different observational data.

Intuitively, handing out the flyer at ‘Enlil Theater’ seems like a good idea – and that’s what a majority of participants did. Both the experimental data and the observational data appear to suggest that the flyer is helping. However, the expected benefit of handing out the flyer is actually negative. In contrast, handing out the flyer at ‘Forseti Theater’ seems like a bad idea – and the majority of participants didn’t hand it out. However, here, the expected benefit is positive.

While the observational data at ‘Enlil Theater’ (Figure 6E) doesn’t rule out any of the counterfactual types, it provides a strong constraint. We can see that the maximum probability of an always-goer is $p(\text{always-goer}) \leq 0.2$ as only four out of twenty bought a ticket. So let’s assume, for now, that $p(\text{always-goer}) = 0.2$ and infer what the rest of the probabilities would need to be. We can infer that $p(\text{complier}) = 0.5$ (using Equation 8), that $p(\text{defier}) = 0.3$ (using Equation 7), and that $p(\text{never-goer}) = 0$ (using Equation 9). While this is one possible solution, maybe there is another one? Let’s assume that $p(\text{always-goer}) = 0.1$. From this we would get that $p(\text{complier}) = 0.6$, which means that $p(\text{defier}) = 0.4$ (using Equation 7). But notice that this doesn’t work. We already have probabilities of these three types that sum to more than 1. The same reasoning holds if we were to try $p(\text{always-goer}) = 0.15$ or $p(\text{always-goer}) = 0$, for example. So we know that the first solution we found is the unique one. Let’s try the same exercise with ‘Forseti Theater’ (Figure 6F). Here, based on the observational data, we know that $p(\text{complier}) \leq 0.2$. Everyone who didn’t take the flyer went (these can’t be compliers), and only four out of ten who took the flyer went (these are either compliers or always-goers). From the fact that the average treatment effect is 0.2, we know that $p(\text{complier}) = 0.2$ and that $p(\text{defier}) = 0$ (using Equation 7). And we can use Equation 8 and Equation 9 to infer that $p(\text{always-goer}) = 0.5$ and $p(\text{never-goer}) = 0.3$.

Overall, what this exercise shows is that it is possible to reason through the different cases to make informed inferences about the probability of the different counterfactual types of people. However, this reasoning doesn’t come naturally to people. Instead, they rely on a simpler strategy that combines the observational and experimental data in a weighted additive way, leaving the experiment with much less money than they could have gotten.

Limitations and future directions

We explored participants’ decisions and inferences in a single context – handing out flyers for movies. Future work should explore different decision contexts. It’s possible, for example, that in our context, participants had strong prior beliefs that flyers are helpful. This might explain why they were overall somewhat biased to hand them out. In other contexts, people’s prior beliefs might be different. Cinelli and Pearl (2021) develop a Bayesian approach that combines causal knowledge of the underlying mechanism with experimental and observational data to make good decisions. While it’s possible that such an approach could help better understand participants’ choices, the fact that their decisions were so well-explained by simple linear weighting suggests that they were relying on more heuristic reasoning.

Future work could also explore a greater range of situations. We focused here on a relatively simple setting that featured binary cause and effect variables, small data sets, and data for which the probabilities of the different counterfactual types can (almost always) be computed exactly, with the same benefit function throughout. Future work may want to

explore how people make decisions with ordinal (Mueller, 2025; Sun, Shi, & Zhao, 2025) or continuous variables (Kawakami, Kuroki, & Tian, 2024; Li & Pearl, 2024). We assumed that our small data sets were a true reflection of the probabilities in the population. One could see whether participants’ decisions are sensitive to sample size manipulations (T. Cheng, Mao, Pearl, & Li, 2026; Dawid, Musio, & Fienberg, 2016), including situations where the observational data isn’t balanced between treatment and non-treatment. It would also be interesting to see how participants perform in more situations where it’s only possible to compute bounds on the counterfactual probabilities. Finally, future work could manipulate the benefit parameters, making it, for example, so that the benefit of giving a flyer to a complier is greater than the harm of giving a flyer to a defier. Remember, however, that we found suboptimal performance in arguably one of the simplest settings. So it’s plausible to assume that people would struggle even more in the extensions we suggested here. That said, future work could explore whether people can learn to make better decisions by receiving minimal instructions that elucidate how the counterfactual calculus works.

Conclusion

Doctors, managers, and policy makers often need to make decisions based on both experimental and observational data. Deciding well requires knowing more than whether an action works on average. It requires knowing whether it would work for the person in front of you – whether they would benefit, be harmed, or be left unaffected. Formal work tells us how to combine the two sources of data to answer that question. But without formal training, people are left to their intuitions. Those intuitions, we found, aren’t up to the task. Even experts in statistics, economics, and data science left a lot of money on the table.

Much work has shown that people rely on heuristics and exhibit various biases when making decisions (Gilovich, Griffin, & Kahneman, 2002; Kahneman, 2011; see also Gigerenzer, Todd, ABC Research Group, et al., 2000). We have uncovered a new heuristic: when presented with experimental and observational data, people combine the available information using a simple additive weighting strategy. However, the correct solution requires a counterfactual calculus that implies a non-linear combination of the available information.

The source of the problem is a failure of inference, not of valuation. Give decision-makers the right beliefs about who they are dealing with, and good decisions follow. Counterfactual reasoning is the bridge from the data to those beliefs – and, as our results suggest, a bridge that has to be taught rather than assumed. For the growing number of stakeholders who hold both experimental and observational data in hand, that training may be what separates a good decision from a costly one.

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Appendix A
Comprehension check questions

This is the set of comprehension check questions that participants needed to answer correctly before being able to proceed to the main part of the experiment. The correct response is bolded.

- The flyer may encourage or discourage people to see the new movie. [**True** / False]
- The effect of the flyer is the same for all movie-goers who frequent the same theater. [True / **False**]
- When the flyers are obtainable, movie-goers are free to choose whether or not to take a flyer. [**True** / False]
- When the flyers are handed out, movie-goers are free to choose whether or not to take a flyer. [True / **False**]
- When you are handing out flyers, movie-goers who don't receive a flyer are still able to obtain one if they want one. [True / **False**]

Appendix B

People inference judgments

Table B1

People inferences for each trial. The first column shows the trial index. The next four columns show the experimental and observational probabilities. The next five columns show the normative probabilities for each of the four types of people as well as the benefit of handing out the flyer. The remaining columns show novices' and experts' inferences together with the benefit computed based on their inferences. Note: c = complier, a = always-goer, n = never-goer, d = defier. Red values indicate negative benefit, and green values positive benefit.

Trial	$p(y_x)$	$p(y_{x'})$	$p(y x)$	$p(y x')$	normative					novice					expert				
					$p(c)$	$p(a)$	$p(n)$	$p(d)$	benefit	$p(c)$	$p(a)$	$p(n)$	$p(d)$	benefit	$p(c)$	$p(a)$	$p(n)$	$p(d)$	benefit
1	0.50	0.50	0.00	0.00	0.50	0.00	0.00	0.50	-5.50	0.18	0.28	0.31	0.23	-3.26	0.10	0.30	0.50	0.09	-1.32
2	0.50	0.50	1.00	1.00	0.50	0.00	0.00	0.50	-5.50	0.32	0.43	0.15	0.10	0.74	0.09	0.55	0.29	0.08	-1.15
3	0.50	0.40	0.00	0.00	0.50	0.00	0.10	0.40	-3.50	0.20	0.21	0.39	0.20	-2.55	0.18	0.23	0.47	0.12	-0.98
4	0.60	0.50	0.20	0.00	0.50	0.10	0.00	0.40	-3.50	0.28	0.23	0.33	0.16	-0.90	0.20	0.33	0.39	0.08	-0.21
5	0.60	0.50	1.00	1.00	0.50	0.10	0.00	0.40	-3.50	0.35	0.43	0.13	0.09	1.34	0.19	0.51	0.23	0.08	-0.21
6	0.60	0.40	0.20	0.00	0.55	0.05	0.05	0.35	-2.00	0.27	0.21	0.33	0.19	-1.58	0.21	0.27	0.41	0.11	-0.55
7	0.70	0.50	0.40	0.00	0.50	0.20	0.00	0.30	-1.50	0.35	0.24	0.28	0.14	0.18	0.25	0.32	0.36	0.07	0.60
8	0.70	0.50	1.00	1.00	0.50	0.20	0.00	0.30	-1.50	0.37	0.41	0.13	0.08	1.60	0.23	0.50	0.20	0.07	0.41
9	0.50	0.30	0.00	0.00	0.50	0.00	0.20	0.30	-1.50	0.22	0.20	0.38	0.19	-2.12	0.23	0.21	0.46	0.10	-0.32
10	0.50	0.50	0.00	1.00	0.00	0.50	0.50	0.00	-0.50	0.17	0.34	0.19	0.30	-4.90	0.07	0.36	0.34	0.23	-4.40
11	0.50	0.50	1.00	0.00	0.00	0.50	0.50	0.00	-0.50	0.43	0.25	0.19	0.13	1.22	0.23	0.35	0.34	0.08	0.26
12	0.50	0.40	1.00	0.00	0.10	0.40	0.50	0.00	0.50	0.47	0.23	0.19	0.11	2.10	0.40	0.25	0.30	0.06	2.32
13	0.60	0.50	0.20	1.00	0.10	0.50	0.40	0.00	0.50	0.24	0.37	0.19	0.21	-2.26	0.18	0.32	0.26	0.24	-3.54
14	0.60	0.50	1.00	0.00	0.10	0.50	0.40	0.00	0.50	0.42	0.25	0.22	0.11	1.41	0.29	0.33	0.31	0.06	1.16
15	0.50	0.20	0.00	0.00	0.50	0.00	0.30	0.20	0.50	0.22	0.19	0.41	0.18	-1.91	0.23	0.18	0.48	0.11	-0.37
16	0.60	0.40	1.00	0.00	0.20	0.40	0.40	0.00	1.50	0.41	0.25	0.22	0.13	1.09	0.33	0.30	0.30	0.07	1.47
17	0.70	0.50	0.40	1.00	0.20	0.50	0.30	0.00	1.50	0.27	0.38	0.19	0.17	-1.20	0.20	0.39	0.23	0.18	-2.15
18	0.70	0.50	1.00	0.00	0.20	0.50	0.30	0.00	1.50	0.43	0.24	0.21	0.13	1.30	0.33	0.32	0.28	0.07	1.42
19	0.50	0.30	1.00	0.00	0.20	0.30	0.50	0.00	1.50	0.42	0.20	0.24	0.15	0.78	0.32	0.26	0.33	0.08	1.18
20	0.70	0.30	1.00	0.60	0.60	0.10	0.10	0.20	1.50	0.43	0.29	0.18	0.10	1.77	0.41	0.30	0.22	0.07	2.32
21	0.50	0.20	1.00	0.00	0.30	0.20	0.50	0.00	2.50	0.41	0.20	0.25	0.14	0.80	0.36	0.21	0.35	0.07	1.65
22	0.70	0.30	1.00	0.00	0.40	0.30	0.30	0.00	3.50	0.48	0.20	0.19	0.13	1.76	0.46	0.24	0.25	0.05	3.01

Appendix C
Performance differences between conditions

Table C1

Posterior estimates of the difference in performance between conditions. The estimate column shows the mean difference in how much bonus participants received with the 95% credible interval. Credible differences are bolded.

comparison	estimate	lower CrI	upper CrI
no feedback – feedback	–0.05	–0.23	0.12
no feedback – data composition	– 0.38	–0.60	–0.16
no feedback – people inference (novice)	– 0.30	–0.51	–0.09
no feedback – people inference (expert)	– 0.80	–1.02	–0.59
feedback – data composition	– 0.33	–0.55	–0.11
feedback – people inference (novice)	– 0.25	–0.46	–0.03
feedback – people inference (expert)	– 0.75	–0.97	–0.54
data composition – people inference (novice)	0.08	–0.18	0.32
data composition – people inference (expert)	– 0.42	–0.69	–0.16
people inference (novice) – people inference (expert)	– 0.50	–0.74	–0.26

Appendix D
Correlations between decisions

Table D1

Correlations between participant decisions across different experiment conditions. Correlations are computed on aggregated responses across trials.

	No feedback	Feedback	Data composition	People inference (novice)
Feedback	.94			
Data composition	.87	.88		
People inference (novice)	.52	.68	.69	
People inference (expert)	.74	.84	.91	.79

Appendix E
Performance in Experiment 3

Table E1

Performance as a function of degree.
Note: The ‘bonus’ column shows means
and 95% credible intervals.

Degree	n	bonus
Undergraduate student	27	1.36 [1.19, 1.52]
Master’s student	1	2.00
PhD student	22	1.26 [1.02, 1.48]
Postdoc	3	1.03 [0.35, 1.80]

Table E2

Performance as a function of familiarity with causal inference (CI). Note: The ‘bonus’ column shows means and 95% credible intervals.

familiarity with CI	n	bonus
not at all	11	1.40 [1.18, 1.62]
slightly	15	1.37 [1.08, 1.64]
somewhat	14	1.12 [0.82, 1.46]
very	13	1.37 [1.15, 1.59]